MS EXCEL

Complete Teacher's Reference Guide

Windows 7 • Office 2007 / 2010

For classroom use — answers to every student question

Topics: Basics · Cells · Formulas · All Functions · Charts · Data Tools

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1. Introduction to MS Excel

What is MS Excel?

Excel is a **spreadsheet program** in Microsoft Office. It stores data in a grid of rows and columns and is mainly used for **calculations, making tables, and drawing charts**.

Think of it this way: An Excel file is like a notebook. Each notebook (Workbook) has pages (Worksheets), and each page has a grid where you type data.

Where Excel is used in real life:

- Student mark-sheets and report cards
- Employee salary sheets
- Shop purchase / sales records
- Monthly budget tracking
- Attendance registers

Version Limits — Important Numbers to Know

Feature	Old Version (2000 / 2003)	New Version (2007 / 2010)
Maximum Rows	65,536	10,48,576
Maximum Columns	256	16,384
File Extension	.xls	.xlsx
Executable File	excel.exe	excel.exe

How to Open Excel in Windows 7

1	Click the **Start** button (bottom-left of screen)
2	Click **All Programs**
3	Click **Microsoft Office**
4	Click **Microsoft Excel 2007** (or 2010)

■ **KEYBOARD SHORTCUTS**

Windows key + R — Opens Run dialog — type excel and press Enter to open Excel directly

STUDENT MIGHT ASK

Q: What is the difference between a Workbook and a Worksheet?

A: A Workbook is the whole Excel file (like a book). A Worksheet is one page inside that file. One workbook can hold up to 256 worksheets.

2. Excel Interface — Key Components

Term	What It Means	Example
Workbook	The entire Excel file you open or save	Sales.xlsx — this is one workbook
Worksheet	One sheet / page inside the workbook	Sheet1, Sheet2, Sheet3
Cell	One box formed where a row meets a column	The box at column B, row 3
Cell Address	Name of a cell: column letter + row number	B3, A1, D10
Active Cell	The cell currently selected — has a dark border	Click on B3 — it becomes the active cell
Cell Pointer	The highlighted dark border around the active cell	Move it using arrow keys or mouse click
Sheet Tab	Labels at bottom of screen showing sheet names — click to switch	Sheet1 | Sheet2 | Sheet3
Formula Bar	Bar at top showing the content / formula of the active cell	Select B3 which has =SUM(A1:A3) — formula bar shows it
Name Box	Small box showing the address of the currently active cell	Shows B3 when cell B3 is selected

EXAMPLE

Cell Address: Column D, Row 5 → Cell Address is D5

If you are in cell A1 and press the right arrow key, you move to B1.

Sheet tabs are at the very bottom of the screen. Right-click a tab to rename it.

3. Working with Cells

3.1 Entering Data into a Cell

Why: Before you can calculate or display anything, you must first type data — numbers, text, or dates — into cells.

1	Click on the cell where you want to enter data (it becomes the Active Cell)
2	Type your text or number
3	Press **Enter** to confirm and move down OR press **Tab** to confirm and move right

EXAMPLE

Click cell B2. Type Ram and press Enter. → Ram is now stored in B2.

Click cell C2. Type 450 and press Tab. → 450 is in C2; cursor moves to D2.

3.2 Editing Existing Cell Content

1	Click on the cell you want to edit
2	**Double-click** the cell to enter edit mode (a cursor appears inside the cell)
3	Make your changes, then press **Enter**

■ KEYBOARD SHORTCUTS

F2 — Press F2 to enter edit mode on the selected cell — same as double-clicking

3.3 Deleting Cell Content

1	Click on the cell (or select a range of cells)
2	Press the **Delete** key on the keyboard

Note: The Delete key removes the *content* but keeps the cell itself. It does NOT remove the cell from the sheet.

3.4 Fill — Copy Content to Nearby Cells Quickly

Why: Instead of typing the same value in 10 cells, type it once and use Fill to copy it instantly in any direction.

Direction	Menu Path	Shortcut
Fill Down (copy downward)	Home tab → Editing group → Fill → Down	Ctrl + D
Fill Right (copy rightward)	Home tab → Editing group → Fill → Right	Ctrl + R
Fill Up (copy upward)	Home tab → Editing group → Fill → Up	—
Fill Left (copy leftward)	Home tab → Editing group → Fill → Left	—

EXAMPLE

Type 500 in cell A1. Select A1:A5. Home → Fill → Down. All cells A1:A5 now show 500.

Shortcut version: Select A1:A5 and press Ctrl + D. Same result, much faster.

4. Navigation — Moving Around the Sheet

Why: Excel sheets can have over a million rows. Knowing how to jump quickly to any cell saves a lot of time.

Key / Shortcut	What It Does
`Arrow Keys (↑ ↓ ← →)`	Move one cell in that direction
`Home`	Jump to the first cell of the current row (e.g. A5 if you are in row 5)
`Ctrl + Home`	Jump to cell A1 — the very beginning of the sheet
`Ctrl + End`	Jump to the last used cell in the sheet
`Ctrl + Arrow Key`	Jump to the last cell with data in that direction (skip blank cells)
`Page Up / Page Down`	Scroll one full screen up or down
`Alt + Page Down`	Move one screen to the right
`Alt + Page Up`	Move one screen to the left
`Ctrl + Page Up`	Move to the previous worksheet (Sheet2 → Sheet1)
`Ctrl + Page Down`	Move to the next worksheet (Sheet1 → Sheet2)
`F5 or Ctrl + G`	Open the Go To dialog — jump to any cell address directly

Using Go To (F5) — Step by Step

1	Press **F5** on the keyboard (Go To dialog box opens)
2	In the **Reference** box, type the cell address you want (e.g. D100)
3	Click **OK** — Excel jumps directly to that cell

EXAMPLE

You are at cell A1 and want to reach cell Z500 instantly.

Press F5 → type Z500 in the Reference box → click OK. Done.

5. Creating Series — Linear, Growth, Date, Custom

What is a Series? A series is a sequence of numbers or dates generated automatically — so you don't have to type every single value.

Why use it? Instead of typing 1, 2, 3 … 100 by hand, Excel fills it in one second using the Fill Series option.

5.1 Linear Series — Numbers increasing by a fixed amount

1	Type the **starting number** (e.g. 1) in a cell and press Enter
2	Select the range of cells where you want the series (e.g. A1:A10)
3	Go to **Home tab** → Editing group → **Fill** → **Series**
4	In the Series dialog: set **Step Value** (how much to add, e.g. 1) and **Stop Value** (last number, e.g. 10)
5	Click **OK**

EXAMPLE

Start: 1 | Step Value: 2 | Stop Value: 9

Result: 1, 3, 5, 7, 9 (adds 2 each time — odd number series)

Start: 5 | Step Value: 5 | Stop Value: 50

Result: 5, 10, 15, 20 ... 50 (5-times table)

5.2 Growth Series — Numbers multiplying by a fixed amount

1	Type the starting number in a cell (e.g. 2 — this is just an example; any number works)
2	Select the range of cells for the series
3	Go to **Home tab** → Editing group → **Fill** → **Series**
4	Select the **Growth** option in the Type section
5	Set Step Value (e.g. 2 to double each time) and Stop Value, then click **OK**

EXAMPLE

Start: 2 | Step Value: 2 | Stop Value: 64

Result: 2, 4, 8, 16, 32, 64 (doubles each time)

Useful for compound interest examples or exponential growth illustrations.

5.3 Date Series — Automatically fill dates

1	Type a starting date in a cell (e.g. 01/04/2014) and press Enter
2	Select the range of cells for the date series

3	Go to **Home tab** → Fill → **Series**
4	Choose type: **Day** (change by 1 day), **Month**, or **Year**
5	Set Step Value (e.g. 1) and Stop Value (last date), then click **OK**

EXAMPLE

Start: 01/04/2014 | Type: Day | Step: 1 | Stop: 07/04/2014

Result: 01/04, 02/04, 03/04, 04/04, 05/04, 06/04, 07/04

Selecting Month gives: 01/04, 01/05, 01/06 ... (first of each month)

5.4 Custom Series — e.g. Monday, Tuesday, Wednesday…

Why: Days of the week and months are already built into Excel. Type Monday and drag down — Excel fills the rest automatically.

1	Click the **Office Button** (round button, top-left)
2	Click **Excel Options**
3	Click **Popular** → then click **Edit Custom Lists**
4	In the List Entries box, type your entries (one per line)
5	Click **Add** → then click **OK**

EXAMPLE

Type Monday in any cell → drag the fill handle (small square at cell corner) downward.

Excel auto-fills: Monday, Tuesday, Wednesday, Thursday ... automatically.

Same works for: January, February, March ... or any list you define.

STUDENT MIGHT ASK

Q: What is the difference between Linear and Growth series?

A: Linear ADDS a fixed number each time (1, 3, 5, 7 — adding 2). Growth MULTIPLIES by a fixed number each time (2, 4, 8, 16 — multiplying by 2).

6. Working with Rows and Columns

6.1 Insert a New Row or Column

Why: If you forgot to add a student's row in the middle of a list, insert a new row — existing rows move down automatically.

1	Click on the row number (or column letter) where you want to insert
2	Go to **Home tab** → Cells group → click **Insert Sheet Rows** (or Insert Sheet Columns)

Faster way: Right-click the row number or column letter → click **Insert**. A new row appears ABOVE the selected row.

6.2 Delete a Row or Column

1	Click on the row number or column letter to select the entire row/column
2	Go to **Home tab** → Cells group → click **Delete Sheet Rows** (or Delete Sheet Columns)

Faster way: Right-click the row number or column letter → click **Delete**.

6.3 Change Row Height and Column Width

Action	Steps	Default Value
Change Row Height	Select row → Home → Cells group → Format → Row Height → type value → OK	15 points (max: 409)
Change Column Width	Select column → Home → Cells group → Format → Column Width → type value → OK	8.43 characters (max: 255)
Auto-fit Row / Column	Double-click the border between two row numbers or between two column letters	Adjusts to fit content

6.4 Freeze Rows and Columns

Why: When you scroll down a long list, the header row disappears. Freeze keeps the header always visible no matter how far you scroll.

1	Click on the cell **below the row** you want to freeze (e.g. click A2 to freeze row 1)
2	Go to **View tab** → Window group → **Freeze Panes** → **Freeze Panes**
3	To unfreeze: View tab → Freeze Panes → **Unfreeze Panes**

EXAMPLE

Row 1 has: Name | Marks | Grade. You have 200 students below.

Click cell A2 → Freeze Panes. Now scroll down — Row 1 always stays visible.

6.5 Hide and Unhide Rows or Columns

Why: Temporarily hide rows or columns you don't need to see right now — data is NOT deleted, just hidden from view.

1	Select the row number(s) or column letter(s) you want to hide
2	Right-click on the selection → click **Hide**
3	The row/column disappears from view but the data is still there

To Unhide: Select the rows/columns on BOTH SIDES of the hidden one (e.g. to unhide row 5, select row 4 and row 6 together) → right-click → **Unhide**.

7. Working with Sheets

Rename Sheet

1	Right-click the sheet tab at the bottom → click **Rename**
2	Type the new name and press Enter
3	Alternative: Double-click the tab directly to rename it

Insert New Sheet

1	Right-click any sheet tab → click **Insert** → select Worksheet → OK
2	Or click the small **+** icon next to the last sheet tab

Delete Sheet

1	Right-click the sheet tab → click **Delete**
2	Click Delete in the warning confirmation box

Warning: Deleted sheets CANNOT be recovered with Ctrl + Z. Be careful.

Hide a Sheet

1	Right-click the sheet tab you want to hide
2	Click **Hide** — the sheet disappears from the tab bar (data is NOT deleted)

A hidden sheet still exists and can be unhidden at any time.

Unhide a Sheet

1	Home tab → Cells group → Format → Hide & Unhide → **Unhide Sheet**
2	Select the sheet name from the list → click **OK**

Protect Sheet (Password)

1	Home tab → Cells group → Format → **Protect Sheet**
2	Type a password in the Password box → click OK
3	Re-type the same password in the Confirm box → click OK

After protection, users cannot edit the sheet without the correct password.

Unprotect Sheet

1	Home tab → Cells group → Format → **Unprotect Sheet**
2	Type the password → click OK

Change Sheet Background

1	Page Layout tab → Page Setup group → **Background**
2	Select an image file → click Insert

Background images do **not** print. They are for on-screen display only.

Remove Background

1	Page Layout tab → Page Setup group → **Delete Background**

STUDENT MIGHT ASK

Q: If I protect a sheet, can anyone still see the data?

A: Yes. Protection only prevents editing. Anyone can still open and read the data. To hide data, hide the sheet (right-click tab → Hide) on top of protection.

8. Operators and Formulas

Key Rule: Every formula in Excel MUST start with an **=** (equals) sign. Without it, Excel treats the entry as plain text.

Why formulas? Instead of manually calculating 40 students' totals, you write one formula and Excel calculates everything instantly — and recalculates automatically whenever data changes.

Arithmetic Operators

Operator	Meaning	Example Formula	Result
`+`	Addition	`=10+5`	15
`-`	Subtraction	`=10-5`	5
`*`	Multiplication	`=10*5`	50
`/`	Division	`=10/5`	2
`%`	Percentage	`=50%`	0.5
`^`	Power / Exponent	`=2^3`	8 (2 to the power 3)

Reference Operators

Operator	Meaning	Example	Explanation
`:(colon)`	Range — selects a block of cells	`=SUM(B1:B10)`	Adds all cells from B1 to B10
`,(comma)`	Union — combines multiple separate ranges	`=SUM(B1:B5, D1:D5)`	Adds B1:B5 AND D1:D5 together

EXAMPLE

=A1+B1 → Adds the values in cells A1 and B1

=A1*B1 → Multiplies value in A1 by value in B1

=SUM(A1:A5) → Adds A1 + A2 + A3 + A4 + A5 all at once

=A1 - B1/2 → Division happens first (B1/2), then subtracted from A1 (BODMAS rule applies)

9. Functions — Math / Arithmetic

What is a Function? A function is a ready-made formula. Instead of writing a long calculation yourself, you call the function by name and give it data — it calculates and returns the answer.

`Syntax: =FUNCTION_NAME( arguments ) e.g. =SUM(A1:A5)`

Function	Syntax	What It Does	Example → Result
SUM	`=SUM(range)`	Adds all numbers in a range	=SUM(10,20,30) → 60
Subtract	`=(A1 - A2)`	Subtracts two values (no SUBTRACT function — use minus sign)	=(50-20) → 30
PRODUCT	`=PRODUCT(n1,n2,...)`	Multiplies numbers together	=PRODUCT(4,3) → 12
ABS	`=ABS(number)`	Absolute value (removes negative sign)	=ABS(-20) → 20
SQRT	`=SQRT(number)`	Square root of a number	=SQRT(49) → 7
FACT	`=FACT(number)`	Factorial (1 × 2 × 3 × … × N)	=FACT(5) → 120
POWER	`=POWER(num,power)`	Raises a number to a given power	=POWER(4,3) → 64 (4×4×4)
LOG	`=LOG(number,base)`	Logarithm (default base = 10)	=LOG(10000) → 4
EVEN	`=EVEN(number)`	Rounds UP to nearest even integer	=EVEN(49) → 50
ODD	`=ODD(number)`	Rounds UP to nearest odd integer	=ODD(50) → 51
ROMAN	`=ROMAN(number)`	Converts Arabic number to Roman numeral	=ROMAN(50) → L
MOD	`=MOD(num,divisor)`	Remainder after division	=MOD(10,3) → 1
ROUND	`=ROUND(num,digits)`	Rounds to specified decimal places	=ROUND(3.567,2) → 3.57
INT	`=INT(number)`	Integer part only (removes decimal)	=INT(7.9) → 7

EXAMPLE

=SUM(B2:B6) → Adds all values in cells B2, B3, B4, B5, B6

=SQRT(144) → Returns 12 (square root of 144)

=FACT(4) → Returns 24 (1 × 2 × 3 × 4 = 24)

=POWER(2,10) → Returns 1024 (2 to the power 10)

10. Functions — Statistical

Why: Use these whenever you need to analyse a set of data — finding average marks, counting students, finding the highest or lowest score, etc.

Function	Syntax	What It Does	Example → Result
AVERAGE	`=AVERAGE(range)`	Mean (average) of numbers	=AVERAGE(20,30,40) → 30
COUNT	`=COUNT(range)`	Counts cells that contain numbers	=COUNT(A1:A10) → number of filled numeric cells
COUNTA	`=COUNTA(range)`	Counts all non-empty cells (text + numbers)	=COUNTA(A1:A5) → 3 (if 3 cells have any content)
MAX	`=MAX(range)`	Highest (maximum) value in range	=MAX(A1:A5) → 95 (highest mark)
MIN	`=MIN(range)`	Lowest (minimum) value in range	=MIN(A1:A5) → 32 (lowest mark)
LARGE	`=LARGE(range,k)`	The k-th largest value	=LARGE(A1:A5,2) → 2nd highest mark
SMALL	`=SMALL(range,k)`	The k-th smallest value	=SMALL(A1:A5,2) → 2nd lowest mark

EXAMPLE

Student marks in C2:C6 → 78, 45, 90, 35, 62

=AVERAGE(C2:C6) → 62 (average mark)

=MAX(C2:C6) → 90 (highest mark)

=MIN(C2:C6) → 35 (lowest mark)

=COUNT(C2:C6) → 5 (5 students have marks entered)

STUDENT MIGHT ASK

Q: What is the difference between COUNT and COUNTA?

A: COUNT only counts cells with NUMBERS. COUNTA counts ALL non-empty cells — including text. Use COUNTA to count student names (which are text).

11. Functions — Logical (IF, AND, OR, NOT)

Why: Logical functions make decisions. They check a condition and take different actions based on whether it is True or False.

Real use: Automatically assign PASS / FAIL, calculate grades, give bonus if a target is met, etc.

11.1 IF Function — the most important logical function

`Syntax: =IF( condition, value_if_true, value_if_false )`

Condition: something that is either True or False (e.g. A1>50)

Value_if_true: what Excel shows if the condition is True

Value_if_false: what Excel shows if the condition is False

EXAMPLE

=IF(C2>=35, "PASS", "FAIL")

→ If marks in C2 are 35 or more: shows PASS. Otherwise: shows FAIL.

Nested IF (multiple conditions — automatic grading):

=IF(C2>=75,"EXCELLENT",IF(C2>=60,"GOOD",IF(C2>=45,"PASS",IF(C2>=30,"PROMOTED","FAIL"))))

Grade conditions: >=75 = EXCELLENT | >=60 = GOOD | >=45 = PASS

>=30 = PROMOTED | below 30 = FAIL

Note (original doc error corrected): The last condition is FAIL for marks BELOW 30,

not >=30. The nested IF naturally falls through to FAIL when no other condition matches.

11.2 AND, OR, NOT Functions

Function	Syntax	Returns TRUE when…	Example
AND	`=AND(c1, c2, ...)`	ALL conditions are True	=AND(A1>10,A1<50) → TRUE only if A1 is between 10 and 50
OR	`=OR(c1, c2, ...)`	ANY ONE condition is True	=OR(A1<0,A1>100) → TRUE if A1 is negative or above 100
NOT	`=NOT(condition)`	Condition is False (reverses True/False)	=NOT(A1>50) → TRUE when A1 is 50 or less

EXAMPLE

AND example — Student eligible only if marks >=35 AND attendance >=75%:

=IF(AND(C2>=35, D2>=75), "ELIGIBLE", "NOT ELIGIBLE")

OR example — Give discount if quantity >100 OR customer is VIP:

=IF(OR(B2>100, C2="VIP"), "DISCOUNT", "NO DISCOUNT")

STUDENT MIGHT ASK

Q: Can I use more than 3 conditions in a nested IF?

A: Yes. Excel 2007 / 2010 supports up to 64 nested IF levels. However, more than 4-5 levels becomes hard to read and debug. For very many conditions consider a VLOOKUP-based lookup table instead.

12. Functions — Date and Time

Why: These functions extract parts of a date or calculate the gap between two dates — useful for age calculation, attendance, project timelines, etc.

Date format in Excel (Windows 7): MM/DD/YYYY — e.g. 08/21/2014 means 21st August 2014.

Function	Syntax	What It Returns	Example → Result
DAY	`=DAY(date)`	Day number (1 to 31)	=DAY("08/21/2014") → 21
MONTH	`=MONTH(date)`	Month number (1 to 12)	=MONTH("02/12/2014") → 2
YEAR	`=YEAR(date)`	4-digit year number	=YEAR("03/05/2014") → 2014
DAYS360	`=DAYS360(start,end)`	Number of days between two dates (360-day year)	=DAYS360("03/02/1996","11/19/2014")
NOW	`=NOW()`	Current date AND time from the system	=NOW() → today's date and current time
TODAY	`=TODAY()`	Current date only (no time)	=TODAY() → today's date, updates daily
WEEKDAY	`=WEEKDAY(date)`	Day of the week as number (1=Sunday…7=Saturday)	=WEEKDAY("04/06/2025") → 1 (Sunday)

EXAMPLE

=TODAY() always shows the current date — updates every time the file is opened.

Approximate age of a student born on 15/08/2000:

=DAYS360("08/15/2000", TODAY()) / 360 → approximate age in years

13. Functions — Financial

Why: Financial functions calculate loan payments, interest rates, and savings growth — without needing manual formulas.

13.1 RATE — Find the interest rate of a loan

```
Syntax: =RATE( nper, pmt, pv )
```

nper = Total number of payments | pmt = Payment per period (enter as negative) | pv = Present Value (loan amount)

EXAMPLE

Loan: Rs. 1,00,000. Paid back in 48 monthly instalments of Rs. 3,000 each.

=RATE(48, -3000, 100000) → Gives the monthly interest rate

Payment is entered as negative because it is money going OUT.

13.2 FV — Future Value of a savings plan

```
Syntax: =FV( rate, nper, pmt, pv, type )
```

rate = interest rate per period | nper = number of periods | pmt = payment per period | pv = starting value | type = 0 (end of period) or 1 (beginning of period)

EXAMPLE

Deposit Rs. 1,000 at the START of every month. Bank pays 12% annual interest.

How much will be saved after 24 months?

=FV(12%/12, 24, -1000, 0, 1) → Total amount after 24 months

12%/12 converts annual rate to monthly. Payment is negative (money going out).

13.3 PMT — Monthly payment on a loan

```
Syntax: =PMT( rate, nper, pv )
```

Calculates how much you must pay each period to clear a loan fully.

EXAMPLE

Loan: Rs. 5,00,000 at 10% annual interest for 5 years (60 months).

=PMT(10%/12, 60, -500000) → Monthly payment amount

14. Functions — Text

Why: Text functions help you clean, combine, split, and transform text data — like joining first and last names, converting case, or counting characters.

Function	Syntax	What It Does	Example → Result
CONCATENATE	`=CONCATENATE(t1,t2,...)`	Joins multiple text strings into one	=CONCATENATE("Ram"," ","Kumar") → Ram Kumar
LEN	`=LEN(text)`	Counts characters (including spaces)	=LEN("HELLO") → 5
UPPER	`=UPPER(text)`	Converts all text to UPPERCASE	=UPPER("hello") → HELLO
LOWER	`=LOWER(text)`	Converts all text to lowercase	=LOWER("HELLO") → hello
PROPER	`=PROPER(text)`	Capitalizes first letter of each word	=PROPER("ram kumar") → Ram Kumar
TRIM	`=TRIM(text)`	Removes extra spaces (keeps single spaces)	=TRIM(" Hello ") → Hello
LEFT	`=LEFT(text,n)`	First n characters from the left	=LEFT("GDCE",2) → GD
RIGHT	`=RIGHT(text,n)`	Last n characters from the right	=RIGHT("GDCE",2) → CE
MID	`=MID(text,start,n)`	n characters starting at position start	=MID("ABCDEF",2,3) → BCD
FIND	`=FIND(find,within)`	Returns position of a character (case-sensitive)	=FIND("D","GDCE") → 2 (D is at position 2)
EXACT	`=EXACT(text1,text2)`	TRUE if strings match exactly (case-sensitive)	=EXACT("Abra","abra") → FALSE (different case)
REPT	`=REPT(text,n)`	Repeats text n times	=REPT("AB",3) → ABABAB
TEXT	`=TEXT(value,format)`	Formats a number as text	=TEXT(1500,"#,##0") → 1,500

EXAMPLE

Names split across cells: A2 = "Ram" B2 = "Kumar"

=CONCATENATE(A2, " ", B2) → Ram Kumar (joined with a space between)

EXACT is case-sensitive: =EXACT("Abra","Abra") → TRUE | =EXACT("Abra","abra") → FALSE

FIND is case-sensitive: =FIND("D","GDCE") → 2 (finds uppercase D at position 2)

Original document errors corrected: (1) EXACT example now shows correct case-sensitive comparison. (2) FIND example uses consistent uppercase. (3) IF condition corrected: the FAIL condition is marks BELOW 30, not '>=30 FAIL' as the original mistakenly listed.

15. Name Range

What is Name Range? Instead of always writing =SUM(B2:B10), you can give that range a name like Marks and write =SUM(Marks). Formulas become much easier to read and understand.

Why use it? When sharing a spreadsheet, named ranges make formulas self-explanatory — no confusion about which cells are being used.

Create a Named Range

1	Select the cell or range of cells you want to name (e.g. B2:B10)
2	Go to **Formulas tab** → Defined Names group → click **Define Name** (dropdown) → click **Define Name**
3	In the New Name dialog, type the name (e.g. Marks) in the Name field
4	Click **OK**

Delete a Named Range

1	Go to **Formulas tab** → Defined Names group → click **Name Manager**
2	Select the name you want to delete from the list
3	Click the **Delete** button → confirm → click **Close**

EXAMPLE

Select cells B2:B6 (student marks). Define Name = Marks.

Now instead of: =SUM(B2:B6) you can write: =SUM(Marks)

Both give the same result — but =SUM(Marks) is far easier to understand at a glance.

16. Conditional Formatting

What is it? Conditional Formatting automatically applies colour, bold, or other formatting to cells based on a condition you set.

Why use it? Instantly highlight failed students (marks < 35) in red, top scorers in green — without manually checking every single cell.

Apply Conditional Formatting

1	Select the cells you want to apply formatting to (e.g. C2:C20 — the marks column)
2	Go to **Home tab** → Style group → click **Conditional Formatting**
3	Click **Highlight Cells Rules** → select a rule type (e.g. Less Than…)
4	Type the value (e.g. 35 for fail condition) in the dialog box
5	Choose the formatting colour from the 'with' dropdown (e.g. Light Red Fill)
6	Click **OK**

EXAMPLE

Select C2:C20 (all student marks).

Conditional Formatting → Highlight Cells Rules → Less Than → type 35 → Red Fill → OK

Every mark below 35 now automatically turns red. No manual checking needed.

You can add multiple rules: <35 = Red | >=35 = Green | >=75 = Dark Green

STUDENT MIGHT ASK

Q: If I change a mark later, does the colour update automatically?

A: Yes. Conditional formatting is live. As soon as you change a value, the formatting updates automatically to match the new value.

17. Cell Alignment and Text Formatting

What is Alignment? Alignment controls where text or numbers sit INSIDE a cell — left, right, center, or justified — both horizontally and vertically.

Why use it? A properly aligned table looks professional and is easier to read. Headers centered, numbers right-aligned, names left-aligned is standard practice.

17.1 Quick Alignment — Using the Home Tab

1	Select the cell(s) whose alignment you want to change
2	Go to **Home tab** → Alignment group
3	Click the desired button: Left, Center, Right, Top Align, Middle Align, Bottom Align, Wrap Text, Merge & Center, or Orientation

Alignment Option	What It Does	Best Used When…
Left Align	Text starts at the left edge of the cell	Names, labels, text data
Center Align	Text is horizontally centered in the cell	Column headers, titles
Right Align	Text sits at the right edge	Numbers, currency, marks
Justify	Text spreads to fill the full cell width	Long paragraphs
Top Align	Content sits at top of cell	When row height is large
Middle Align	Content is vertically centered	Standard for most tables
Bottom Align	Content sits at bottom of cell	When row height is large
Wrap Text	Long text breaks into multiple lines inside cell	Text too long for one line
Merge & Center	Combines multiple cells into one wide cell	Titles spanning multiple columns
Orientation	Rotates text to an angle (e.g. 45 degrees)	Narrow column headers
Shrink to Fit	Reduces font size automatically to fit in cell	Cannot widen the column

17.2 Full Alignment Options — Format Cells Dialog

1	Select the cell(s) to format
2	Press **Ctrl + 1** (or Home tab → click the small arrow in the Alignment group corner)
3	The Format Cells dialog opens — click the **Alignment** tab
4	Set Horizontal alignment (General, Left, Center, Right, Justify)
5	Set Vertical alignment (Top, Center, Bottom, Justify)

6	Check **Wrap text** to allow text to break into multiple lines inside the cell
7	Check **Merge cells** to combine the selected cells into one
8	Set **Orientation** to rotate text (drag the dial or type degrees)
9	Click **OK**

■ KEYBOARD SHORTCUTS

Ctrl + 1 — Opens Format Cells dialog directly from anywhere

EXAMPLE

Make a title span columns A to E:

Select A1:E1 → Home tab → Alignment group → Merge & Center.

The 5 cells merge into one wide cell. Title is centered across all columns.

Fix text that is too long for a cell:

Select the cell → Home tab → Alignment group → Wrap Text.

Text wraps to multiple lines inside the cell. Row height adjusts automatically.

18. Apply Border to Cell

What is a Border? The grey grid lines you see in Excel do NOT print by default. Borders are actual lines you add to cells that print and display exactly as you set them.

Why use it? Borders make tables look professional and clear. You can add thick outer borders, thin inner lines, coloured lines, or dotted lines.

18.1 Quick Border — Using the Home Tab Border Button

1	Select the cells where you want to add a border
2	Go to **Home tab** → Font group
3	Click the **dropdown arrow** next to the Border button (the box icon with lines)
4	Select your border style: All Borders, Outside Borders, Thick Box Border, etc.

18.2 Custom Border — Full Control via Format Cells Dialog

1	Select the cells to add border to
2	Press **Ctrl + 1** to open Format Cells dialog
3	Click the **Border** tab
4	Choose a **Line Style** from the Style panel (solid, dotted, dashed, double, etc.)
5	Choose a **Color** from the color dropdown
6	Click border preset buttons (**Outline** for outer border, **Inside** for inner grid) OR click directly on the preview diagram to add or remove individual border lines
7	Click **OK**

Border Option	What It Applies
None	Removes ALL borders from selected cells
Outline	Border around the OUTSIDE edge of the entire selection only
Inside	Borders between all cells INSIDE the selection (inner gridlines)
All Borders	Border around every individual cell in the selection
Thick Box Border	Thick outer border — great for highlighting total rows or headers
Bottom Border	Line at bottom of cells only — like an underline
Top and Bottom	Lines at both top and bottom — used for total rows in accounts

EXAMPLE

To make a marks table look professional:

Step 1: Select entire table A1:D10 → Ctrl+1 → Border → All Borders → OK.

Step 2: Select only header row A1:D1 → Ctrl+1 → Border → choose Thick line style → click Outline → OK.

Result: all cells have thin inner borders, header row has a thick outer border.

STUDENT MIGHT ASK

Q: Why don't the grid lines show when I print?

A: The light grey lines on screen are just visual guides — they don't print by default. To print gridlines: Page Layout tab → Sheet Options → check 'Print' under Gridlines. OR add Borders to cells you want lines on — borders always print.

19. Charts

What is a Chart? A chart is a visual / graphical representation of data in your worksheet — bar graph, pie chart, line graph, etc.

Why use it? Numbers in a table are hard to compare visually. A bar chart showing 5 students' marks lets you see the best and worst at a glance.

Create a Chart

1	Select the data you want to visualise (e.g. A1:B6 — names and marks)
2	Go to **Insert tab** → Charts group
3	Click your desired chart type: Column, Bar, Line, Pie, etc.
4	The chart is inserted into the current worksheet automatically

Chart Type	Best Used For
Column / Bar Chart	Comparing values across categories (e.g. marks of different students)
Line Chart	Showing trends over time (e.g. monthly sales growth)
Pie Chart	Showing parts of a whole (e.g. % of pass / fail students)
Area Chart	Like a line chart but with filled area (shows volume over time)
Scatter / XY	Showing relationship between two variables

Delete a Chart

1	Click once on the chart to select it
2	Press the **Delete** key on the keyboard

EXAMPLE

Select A1:B6 where A = Student Names and B = Marks.

Insert → Column Chart. A bar graph appears comparing each student's marks.

If you update any mark in the sheet, the chart updates automatically.

20. Sort and Filter

20.1 Sort — Arrange Data in Order

Why: Sort a student list alphabetically (A to Z) or sort marks from highest to lowest. The entire row moves together — data is never mismatched.

1	Click on any cell within the column you want to sort by
2	Go to **Data tab** → Sort & Filter group
3	Click **A to Z** for ascending order (A, B, C… or 1, 2, 3…)
4	Click **Z to A** for descending order (Z, Y, X… or 100, 90, 80…)

EXAMPLE

Click any cell in the Marks column. Data tab → Z to A.

Entire table re-arranges with highest mark at top. All other columns move with it.

20.2 Filter — Show Only Specific Rows

Why: From a list of 500 students, show only those who scored below 35 (failed) — without deleting others. Filter hides the rest temporarily.

1	Click any cell within your data
2	Go to **Data tab** → Sort & Filter group → click **Filter**
3	Dropdown arrows (▼) appear on every column header
4	Click the arrow on the column you want to filter → select values or use Number Filters
5	To remove the filter: Data tab → click **Filter** again (toggles off)

EXAMPLE

Click Filter. Click the ▼ on the City column. Uncheck all cities except Mumbai.

Only Mumbai students are shown. Others are hidden (not deleted).

Remove filter: Data tab → Filter → all rows reappear.

STUDENT MIGHT ASK

Q: Does filtering delete the hidden rows?

A: No. Filtered rows are only HIDDEN from view — not deleted. Remove the filter and all rows reappear. This is just a way to temporarily focus on specific data.

21. Subtotal

What is Subtotal? Subtotal groups your data and calculates a summary (sum, average, count, etc.) for each group automatically.

Why use it? From a sales list, sum Pens separately, Pencils separately, Charts separately — all in one step.

Important — Sort First: Before using Subtotal, you MUST sort the data by the grouping column. (e.g. sort by Item Name so all Pens are together, all Pencils together.) Without sorting, groups will be calculated incorrectly.

1	**First:** Sort the data by the column you want to group by (e.g. sort by Item Name)
2	Click any cell within the data
3	Go to **Data tab** → Outline group → click **Subtotal**
4	In the Subtotal dialog: set 'At each change in' = the group column (e.g. Item Name)
5	'Use function' = select Sum (or Average, Count, Max, Min)
6	Check the column to calculate in 'Add subtotal to' (e.g. Total Price)
7	Click **OK**

Remove Subtotal

1	Place cursor within the subtotal data
2	Data tab → Outline group → **Subtotal** → click **Remove All**

EXAMPLE

Item data: Pen ×3, Pen ×2, Pencil ×5, Pencil ×2, Marker ×1

After sorting by Item Name → Subtotal → At each change in: Item Name → Sum → Total Price

Result: Pen Total: xxx | Pencil Total: xxx | Marker Total: xxx

22. Data Validation

What is it? Data Validation restricts what a user can type in a cell. You set rules — only numbers 0 to 100 allowed, only dates after 2020, etc.

Why use it? Prevents mistakes. A student's marks cannot be 500 or -10. Validation shows an error if someone tries to enter an invalid value.

Example: Restrict marks to whole numbers 0 to 100 only

1	Select the cells where marks will be entered (e.g. C2:C50)
2	Go to **Data tab** → Data Tools group → click **Data Validation**
3	Settings tab: Allow = **Whole Number** | Data = **between** | Minimum = 0 | Maximum = 100
4	Click the **Input Message** tab → type a helpful hint (e.g. 'Enter marks between 0 and 100')
5	Click the **Error Alert** tab → type an error message (e.g. 'Invalid! Marks must be between 0 and 100')
6	Click **OK**

EXAMPLE

Cell C2 now only accepts whole numbers between 0 and 100.

If someone types 150 or -5, an error message pops up and the entry is rejected.

The input message appears as a tooltip when the cell is clicked — guiding the user.

Remove Validation

1	Select the validated cells
2	Data tab → Data Validation → Settings tab → click **Clear All** → OK

STUDENT MIGHT ASK

Q: Can I create a dropdown list using Data Validation?

A: Yes. In the Allow box, choose 'List' instead of Whole Number. Then type the options separated by commas (e.g. Pass,Fail,Promoted). The cell gets a dropdown arrow.

23. Goal Seek

What is Goal Seek? Goal Seek works backwards. You tell Excel what result you want — and it finds the input value needed to get there.

Analogy: You want to score 75% overall. You know marks in 4 subjects. Goal Seek tells you exactly what you need in the 5th subject.

Setup for this example (Simple Interest):

A1 = Principle (e.g. 900) | B1 = Rate of Interest (e.g. 5) | C1 = Time in years (e.g. 5)

`D1 = Formula: =(A1*B1*C1)/100 → gives Simple Interest = 225`

Use Goal Seek to find what Rate gives Interest = 400

1	Place cursor on the **formula cell** (D1 — the Interest result cell)
2	Go to **Data tab** → Data Tools group → **What-If Analysis** → **Goal Seek**
3	Set Cell: D1 (the cell with the formula)
4	To Value: 400 (the result you want to achieve)
5	By Changing Cell: B1 (the Rate cell — the unknown value you want Excel to find)
6	Click **OK** — Excel fills in the Rate needed to produce Interest = 400

EXAMPLE

Before Goal Seek: Rate = 5% → Interest = 225

After Goal Seek (To Value = 400): Excel changes Rate to ~8.89% → Interest = 400

Very useful: you know the TARGET, and Goal Seek finds the INPUT for you.

24. Scenario Manager

What is Scenario Manager? It lets you save and compare multiple different versions of your data (called scenarios) and switch between them instantly.

Real example: Best case (high sales), Worst case (low sales), Average case — save all three and compare side by side in a summary report.

1	Go to **Data tab** → Data Tools group → **What-If Analysis** → **Scenario Manager**
2	Click **Add** button
3	Type a Scenario Name (e.g. Best Case) → click OK
4	In Scenario Values dialog, type the values for the changing cells → click OK
5	Repeat steps 2–4 to add more scenarios (e.g. Worst Case, Average Case)
6	To view a scenario: select it from the list → click **Show**
7	To compare all: click **Summary** → select Scenario Summary → click OK

EXAMPLE

Scenario 1 (Best Case): Sales = 1,00,000 Expenses = 30,000

Scenario 2 (Worst Case): Sales = 40,000 Expenses = 35,000

Scenario 3 (Average): Sales = 70,000 Expenses = 32,000

Summary report shows all three scenarios side-by-side on a new sheet automatically.

25. Formula Auditing

What is Formula Auditing? It shows visual arrows on the worksheet to trace which cells a formula uses (Precedents) and which formulas depend on a selected cell (Dependents). Helps you find errors.

Why use it? When a formula gives a wrong result, auditing shows exactly which cells are feeding into it — so you can find and fix the mistake.

Feature	What It Does	How to Use
Trace Precedents	Shows arrows pointing TO the formula cell — reveals which cells it uses	Select formula cell → Formulas tab → Trace Precedents
Trace Dependents	Shows arrows FROM a cell — reveals which formulas USE this cell	Select cell → Formulas tab → Trace Dependents
Remove Arrows	Clears all auditing arrows from the worksheet	Formulas tab → Formula Auditing group → Remove Arrows
Show Formulas	Displays actual formula text in cells instead of calculated values	Formulas tab → Show Formulas OR press Ctrl + `
Error Checking	Scans for formula errors like #DIV/0!, #VALUE!, #REF!	Formulas tab → Formula Auditing group → Error Checking

EXAMPLE

Cell D2 has formula =A2+B2+C2 but shows the wrong result.

Click D2 → Formulas tab → Trace Precedents.

Blue arrows appear from A2, B2, C2 pointing to D2.

Verify visually that the formula is using the correct source cells.

26. Pivot Table

What is a Pivot Table? A Pivot Table is an interactive summary that lets you reorganise and summarise large amounts of data with just a few clicks — no formulas needed.

Why use it? From a list of 1000 sales records, instantly find: total sales by city, average marks by class, count of orders per product, etc.

Think of it as: A smart summary report that you can rotate, filter, and reorganise interactively.

1	Select any cell within your data table
2	Go to **Insert tab** → Tables group → click **PivotTable**
3	The Create PivotTable dialog appears — click **OK** (default settings are fine)
4	A new sheet opens with the **PivotTable Field List** panel on the right
5	Drag field names into the 4 areas: Row Labels, Column Labels, Values, Report Filter
6	The Pivot Table builds itself automatically as you drag fields

Drop Area	Purpose	Example
Row Labels	Each unique value becomes one row	Student names as rows
Column Labels	Each unique value becomes one column header	Subjects as columns
Values	The calculation area (Sum, Average, Count…)	Sum of marks per student
Report Filter	A master filter for the entire table	Filter by Class — show Class 10 only

EXAMPLE

Sales sheet: Salesperson | Product | Month | Amount

Drag Salesperson → Row Labels. Drag Month → Column Labels. Drag Amount → Values (Sum).

Instantly: each row = one salesperson, each column = one month, each cell = total sales for that person in that month.

Change Values from Sum to Average → table immediately shows average sales instead.

STUDENT MIGHT ASK

Q: If the source data changes, does the Pivot Table update automatically?

A: No. You must right-click the Pivot Table and click 'Refresh' to update it with new data. It does NOT update automatically when source data changes.

Quick Keyboard Shortcuts Reference

Shortcut	Action
`Ctrl + S`	Save the file
`Ctrl + Z`	Undo last action
`Ctrl + Y`	Redo
`Ctrl + C`	Copy selected cells
`Ctrl + X`	Cut selected cells
`Ctrl + V`	Paste
`Ctrl + A`	Select all cells
`Ctrl + D`	Fill Down
`Ctrl + R`	Fill Right
`Ctrl + Home`	Jump to cell A1
`Ctrl + End`	Jump to last used cell
`Ctrl + G / F5`	Open Go To dialog — jump to any cell address
`F2`	Edit the selected cell
`F4`	Repeat last action / toggle absolute reference (A1)
`Delete`	Clear cell content
`Ctrl + 1`	Open Format Cells dialog
``Ctrl + ` ``	Toggle between showing values and showing formulas
`Ctrl + Page Down`	Move to next worksheet
`Ctrl + Page Up`	Move to previous worksheet
`Alt + Enter`	Start a new line inside the same cell
`Ctrl + Shift + L`	Toggle AutoFilter on / off
`Ctrl + T`	Convert data range to a formal Table